

Reproducing Yokum, Ravishankar and Coppock (2019)

A randomized control trial evaluating the effects of police body-worn cameras

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This repository reproduces the published results of Yokum, Ravishankar and Coppock (2019), a randomized trial of police body-worn cameras conducted with the Metropolitan Police Department of the District of Columbia. It contains a maintained rewrite of the analysis in current R, a table that compares every checkable published number against both the deposited code and the rewrite, and a report on what does and does not reproduce.

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0.1 What is here

Table 1: Where the paper and its materials live. Both DOIs were resolved and checked against the registration agency's record.

Item	Reference
Article	https://doi.org/10.1073/pnas.1814773116
Supplementary appendix	https://www.pnas.org/lookup/suppl/doi:10.1073/pnas.1814773116/-/DCSupplemental
Deposited archive	https://doi.org/10.17605/OSF.IO/P6VUH
Preanalysis plan	Included in the deposited archive and reprinted as Appendix G

yokum_ravishankar_coppock_2019/

README.qmd	this report, rendered to README.md and report/
yokum_ravishankar_coppock_2019.Rproj	
run_all.R	entry point: runs everything in order
download_original.R	fetches and verifies the deposited archive
original_manifest.csv	the deposit's file names, sizes and checksums
original/	the deposited archive (not redistributed here)
maintained/	the rewrite, one script per published float
maintained/output/	everything the rewrite produces
ground_truth/	published values against archive and rewrite output
report/	the PDF build of this report

To reproduce: clone the repository, open yokum_ravishankar_coppock_2019.Rproj, and run

```
source("run_all.R")
```

The first run downloads the deposited archive from OSF, which is 176 MB, and verifies every file against the checksums OSF publishes. Later runs skip files already present and correct. Each script under `maintained/` can also be run on its own.

1 Summary

Does the deposited archive run? Yes. All four deposited scripts execute at exit status 0 under R 4.6.0 and current package versions. The only complaints are a deprecation warning and a layout warning, both from the two coefficient plot scripts.

Does the deposited archive reproduce the paper? Almost none of it. The archive's one table-producing script fits every model on all 2,224 officers in the trial. The published

tables report 1,922, the officers in the seven patrol districts, which is the sample the article says in its Methods section it analyses. Of the 205 published table values the archive’s own output can be compared against, 189 disagree and 16 agree, and every one of the 16 agreements is a cell whose published value is zero or a standard error that happens to round the same way. `regression_tables.R` produces those numbers with no error, no warning and nothing on screen to suggest a problem.

Does the maintained rewrite reproduce the paper? Yes, on everything the deposit can support. Of 264 checkable published values, 263 agree to the precision the paper prints and 1 does not. The single disagreement is an in-text number that disagrees with the paper’s own appendix table. A further 30 rows record published quantities that no deposit could reproduce, almost all of them because the officer covariates behind the covariate-adjusted results cannot be published without identifying individual officers.

2 The paper

Yokum, David, Anita Ravishankar and Alexander Coppock. 2019. “A randomized control trial evaluating the effects of police body-worn cameras.” *Proceedings of the National Academy of Sciences* 116 (21): 10329 to 10332.

Roughly half of 2,224 eligible Metropolitan Police Department officers were randomly assigned a body-worn camera, blocked on district and on officer characteristics within district. Outcomes came from administrative records over matched 212-day windows before and after each district’s camera rollout, and were expressed as yearly rates per 1,000 officers. The paper reports two estimators: a difference in means weighted by the inverse probability of assignment, and a regression that adds the pretreatment value of the outcome and officer covariates. The main analysis is the seven patrol districts, 1,922 officers, excluding the station detail and the special units.

One small discrepancy inside the paper is worth noting before anything is compared to it. Eq. 1 in the Methods section writes the covariate-adjusted specification with a vector of major-block indicators alongside the pretreatment outcome and the officer covariates. None of the sixteen published covariate-adjusted tables lists a block indicator: each prints the pretreatment outcome, gender, a three-category race variable and length of service, and nothing else. The inverse probability weights already account for the differing assignment probabilities across blocks, so the omission is defensible, but the equation and the tables do not describe the same regression.

The headline result is a null. Across all 45 measured outcomes, covering use of force, civilian complaints, policing activity and judicial outcomes, cameras changed nothing detectably.

3 The deposited archive

The deposit at osf.io/p6vuh holds 11 files in two folders, 176 MB in total.

Table 2: The deposit, as listed by the OSF API.

File	Bytes	OSF id
Pre-Analysis Plan (PAP)/Pre-analysis Plan - MPD BWC RCT - v1.pdf	408,266	hpmrt
Replication Data/BWC_dv_df.rdata	1,855	kmjxg
Replication Data/README.txt	593	4pzyt
Replication Data/coefplot_all_dvs.R	1,914	vswpu
Replication Data/coefplot_main_dvs.R	1,920	f2c93
Replication Data/day_level_anon.csv	177,902,506	h9wpc
Replication Data/dv_df.csv	8,985	ctyb9
Replication Data/officer_level_anon.csv	1,865,518	gsbwt
Replication Data/regression_tables.R	15,972	muqpj
Replication Data/time_series_plot.R	3,292	spr5g
TheLabDC_MPD_BWC_Working_Paper_10.20.17.pdf	4,650,695	rk2pa

`download_original.R` fetches each file by its OSF identifier and checks it against both published hashes. All 11 files verify on MD5 and on SHA-256, and the local copy is byte for byte what OSF serves. The script also refuses to continue if `original/` holds anything the manifest does not list, so a stray file left behind by a previous run cannot quietly become part of the archive.

Two notes on the shape of the deposit. It carries a preprint of the paper from 2017, not the published article, so the published values used below were read from the article and its appendix rather than from anything in the deposit. And the deposit's own `README.txt` lists five R scripts where four are present: `regression_tables_cov.R`, the script that would have produced the covariate-adjusted tables, is named but absent.

3.1 What the deposited scripts do

Table 3: The deposited scripts and the published floats they correspond to.

Script	Produces	Status
<code>coefplot_main_dvs.R</code>	Figure 1	Runs, two warnings
<code>coefplot_all_dvs.R</code>	Figure C.1	Runs, nine warnings
<code>regression_tables.R</code>	Tables C.5 to C.37, odd numbered	Runs clean
<code>time_series_plot.R</code>	Figures E.3 and E.4	Runs clean

Everything that is present runs, all four at exit status 0. `coefplot_main_dvs.R` and `coefplot_all_dvs.R` raise a deprecation warning for `geom_errorbarh()` and then one layout warning from `position_dodgev()` per panel drawn, which is one warning for the single-panel figure and eight for the faceted one. Nothing errors. No script saves an output either: all four draw to the screen or print to the console, so a reader who runs them has nothing on disk to compare against anything. Run non-interactively each one leaves an `Rplots.pdf` in the working directory, which is the only reason to care where they are run from, and it is why the archive here is copied out before it is run rather than executed inside `original/`.

3.2 Three ways the deposited code departs from the paper

The sample. Every model in `regression_tables.R` is fit on the full 2,224 officers. Every published table reports 1,922. The two coefficient plot scripts have the same problem. This single difference is why the deposited output does not match the paper anywhere, and it is invisible at runtime.

Table 4: Table C.5, the first of the appendix tables. Standard errors in parentheses. The deposited script’s numbers come from all 2,224 officers; the published table and the rewrite use the 1,922 in the seven patrol districts.

Outcome	Published	Deposited script	Rewrite	N, deposited	N, rewrite
Use of Force	73.6 (87.0)	69.8 (76.6)	73.6 (87.0)	2224	1922
Use of Force (Serious)	13.8 (14.1)	8.4 (12.7)	13.8 (14.1)	2224	1922
Use of Force (Other)	59.8 (83.4)	61.4 (73.4)	59.8 (83.4)	2224	1922

The second estimator in the figures. Both coefficient plot scripts build a vector of covariate-adjusted formulas and then never use it: the object named for the adjusted fits is computed from the unadjusted formulas instead. The figures the deposited scripts draw therefore plot the same estimates twice, one series labelled as though it were adjusted. The rewrite uses the adjusted formulas, which is plainly what the deposited code intends.

The adjustment itself. Even with that corrected, the adjusted series cannot be the published one. Every published covariate-adjusted table conditions on officer gender, race and length of service, and the deposit contains none of the three, and says so, because releasing them would identify individual officers. The deposited scripts substitute an adjustment on the pretreatment outcome alone and note the substitution in a comment. The rewrite keeps that reduced adjustment and labels it for what it is. No published number from any covariate-adjusted table or figure series can be checked against this deposit, by anyone.

4 The maintained rewrite

`maintained/` holds one script per published float, a shared `helpers.R` that defines the two estimators and the analysis sample, and two cleaning scripts. Output goes to `maintained/output/` and is committed, so a fresh run can be diffed against it without downloading anything.

Table 5: The rewrite.

Script	Produces
<code>helpers.R</code>	packages, paths, the analysis sample, the two estimators
<code>clean_officer_level.R</code>	the officer file, whole and cut to the seven districts
<code>clean_day_level.R</code>	the officer-day panel collapsed to district by period rates
<code>table_a1_random_assignment.R</code>	Table A.1
<code>table_a4_deployment_dates.R</code>	Table A.4
<code>table_c5 ... table_c37</code> (17 scripts)	the odd-numbered appendix tables, C.5 through C.37
<code>figure_1_main_outcomes.R</code>	Figure 1
<code>figure_c1_all_outcomes.R</code>	Figure C.1
<code>figure_e3_use_of_force_time_series.R</code>	Figure E.3
<code>figure_e4_complaints_time_series.R</code>	Figure E.4
<code>text_main_estimates.R</code>	the three effect estimates the Results section states in words
<code>text_sample_and_compliance.R</code>	the sample counts and the compliance means in Methods
<code>text_null_results_check.R</code>	a count of outcomes reaching significance
<code>text_district_by_district.R</code>	the seven district-level estimates the Discussion asserts

Substitutions from the deposited code: `%>%` becomes the native pipe; `geom_errorbarh()` becomes `geom_linerange()`; `position_dodgev()` from the unmaintained `coefplot` package becomes `position_dodge()` with the outcome on the discrete axis; `lm()` with `starprep()` becomes `lm_robust()`, whose default standard error is already HC2; `stargazer()` printing LaTeX to the console becomes a CSV at full precision plus a rounded LaTeX table written to a file; and `rm(list = ls())` at the top of every script is gone, since the scripts assume nothing about the session they start in.

Two smaller decisions are worth recording. The weights column is pulled out of the data frame before it reaches `lm_robust()`, because a weights argument is evaluated inside the data first and the column is called `weights`. And district labels are read from CSV as text on purpose: `readr` accepts the Fortran double exponent, so a guessed column of 1D through 7D comes back as the numbers 1 through 7.

4.1 Coverage

Every published float has at least one row in the ground truth. Five are recorded there as having no source in the deposit at all:

- **Tables A.2 and A.3**, the pretreatment distributions of officer race, sex and length of service. Those covariates are not deposited.
- **Tables C.6 through C.36, even numbered**, the sixteen covariate-adjusted tables. Same reason, and the script that made them is missing besides.
- **Figure D.2**, all outcomes under the alternate measurement strategy, which uses each district’s full observation window. The deposited officer file carries only the 212-day rates.
- **Figure E.5**, calls for service against videos uploaded. The calls-for-service series is not deposited.
- The **adherence figures** in the Discussion, 98 per cent of days and 96 per cent average adherence, which rest on the same calls-for-service data.

5 Ground truth

`ground_truth/build_ground_truth.R` assembles 294 rows. Each carries the published value, typed from the article or the appendix and used only as a comparison target; the value the deposited specification produces, read from a file written by `ground_truth/archive_estimates.R`; and the value the rewrite produces, read from `maintained/output/`. No published number is an input to any computation anywhere in this repository.

Table 6: Ground truth, all rows. A row is not checkable when the paper states no number, or when nothing in the deposit can produce one.

Comparison	Agree	Disagree	Not checkable
Deposited specification against the paper	16	189	89
Maintained rewrite against the paper	263	1	30

Table 7: Checkable rows by where the number appears.

Published float	Rows	Rewrite agrees	Rewrite disagrees
Appendix Tables C.5 to C.37	205	205	0
Appendix Tables A.1 and A.4	43	43	0
In-text quantities	16	15	1

The one disagreement is the compliance sentence in Methods. The article says treatment officers uploaded “about 665” videos a year. The weighted mean in the analysis sample is 663.1, which is also what the article’s own Table C.37 implies, since its constant of 13.9 plus its coefficient of 649.1 is 663.0. No sample or weighting scheme available in the deposit returns 665. The gap is two videos out of 663 and changes nothing, but the sentence and the table it summarizes do not agree with each other.

Two checks in the ground truth are of claims the paper makes in words and supports with no table.

Table 8: Claims stated in prose and checked against the data.

Claim	Paper implies	Rewrite finds
Outcomes significant at conventional levels	0	0
Districts with a significant effect on use of force	0	0

Both hold. Across the 45 outcomes, no unadjusted estimate reaches $p < 0.10$, let alone $p < 0.05$; two outcomes, serious uses of force against white and against other-race civilians, are identically zero throughout the posttreatment window and have no p-value at all. District by district, no estimate on use of force is significant at the 0.05 level.

6 Figures

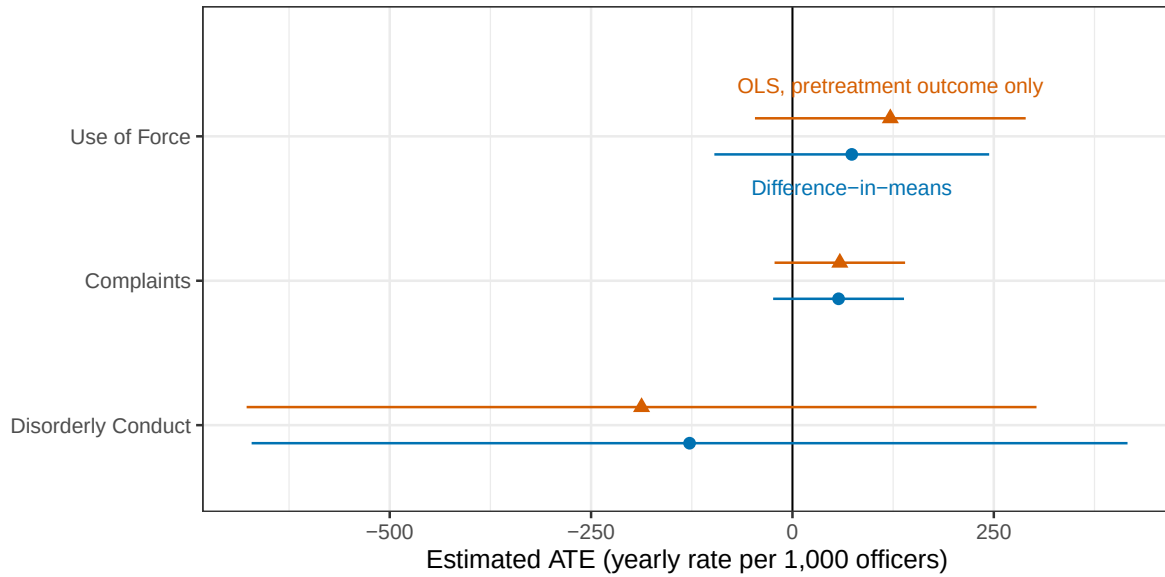


Figure 1: Figure 1 of the article. The circles are the published difference-in-means estimates. The triangles are not the published second series: the deposit cannot support the covariate adjustment the paper used, so this adjusts for the pretreatment outcome alone.

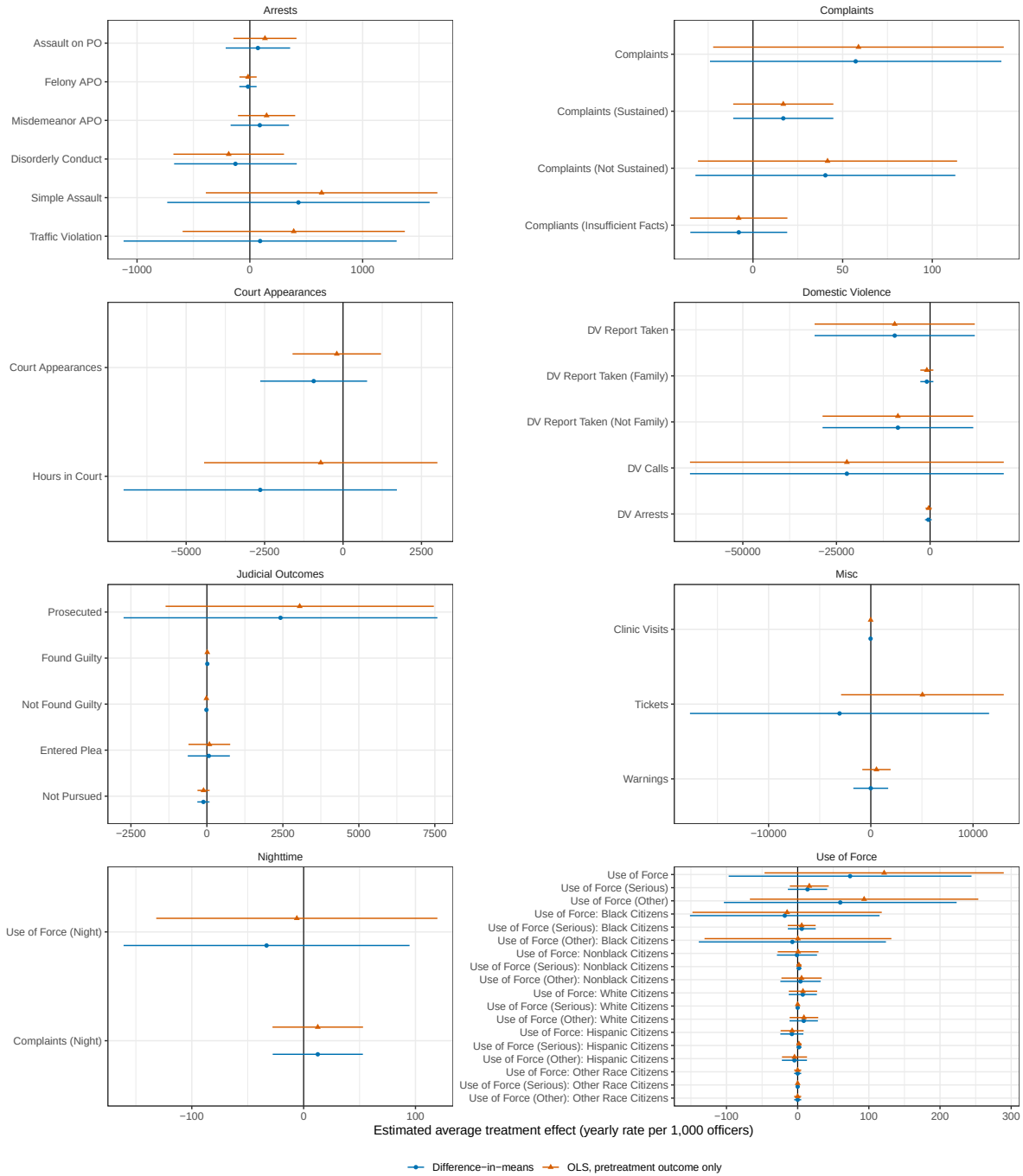


Figure 2: Appendix Figure C.1, all 45 outcomes. The same caveat applies to the adjusted series.

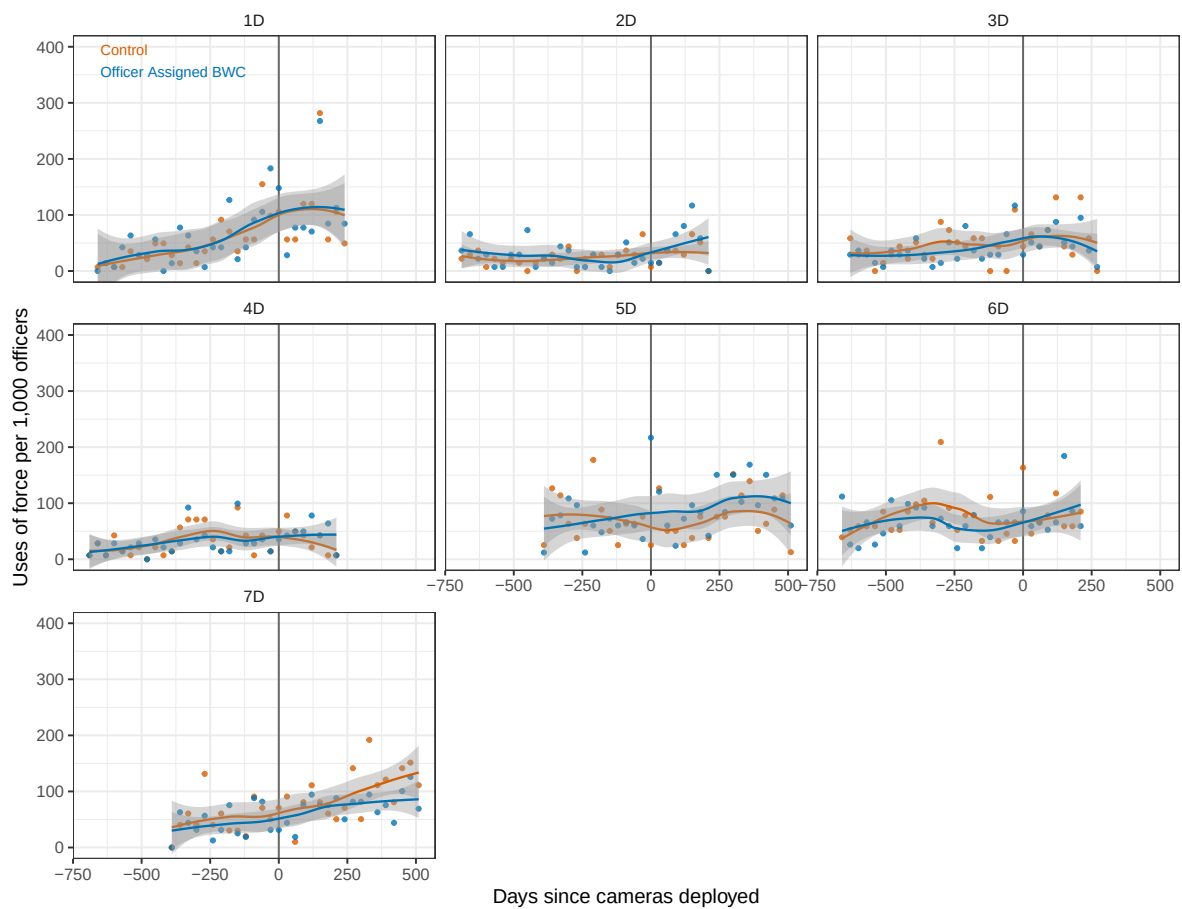


Figure 3: Appendix Figure E.3, uses of force per 1,000 officers by district and 30-day period.

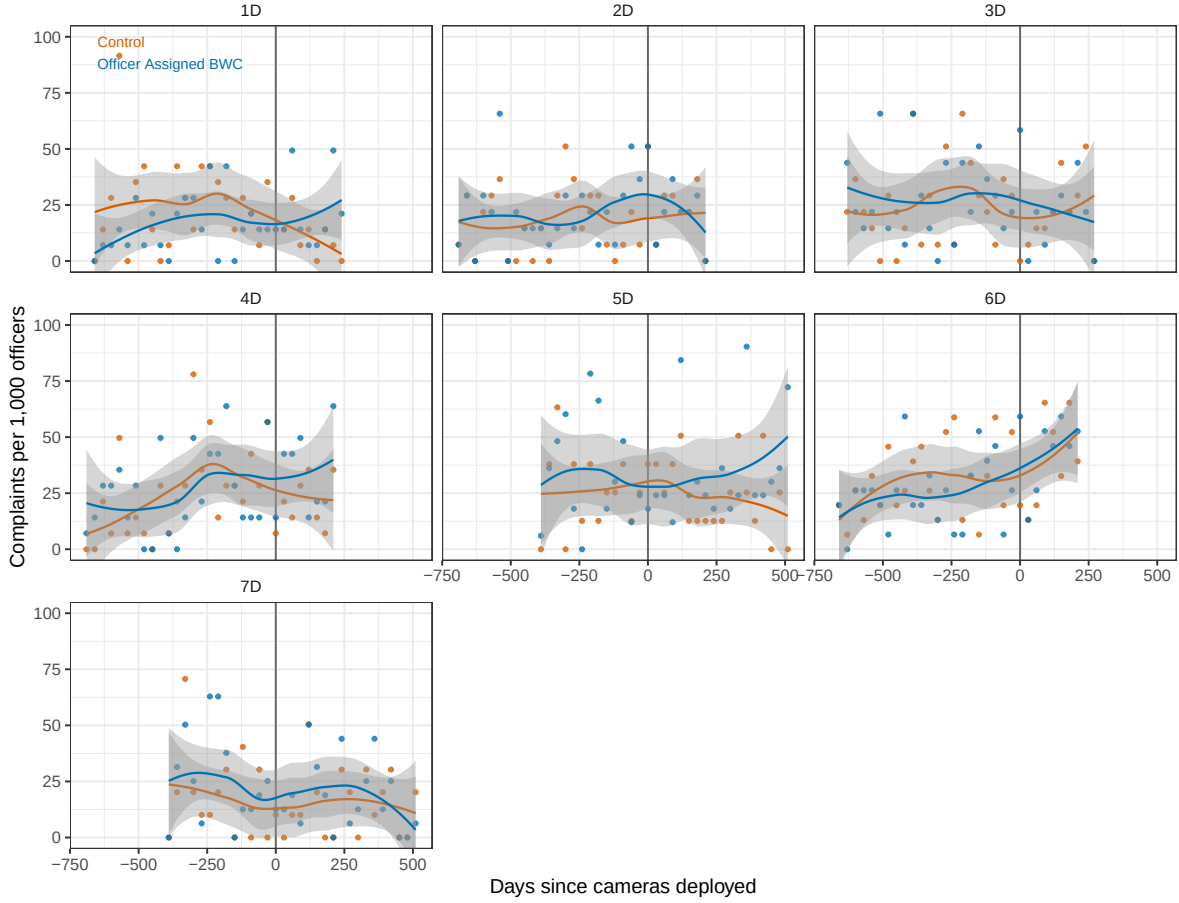


Figure 4: Appendix Figure E.4, complaints per 1,000 officers by district and 30-day period.

None of the published figures prints a number, so each figure script writes a CSV of the values it plots and the ground truth compares those. Figure C.1 plots the same unadjusted estimates the appendix tables print, and across 45 outcomes, matched on the outcome column name rather than the printed label, the largest absolute difference between the figure's values and the tables' is 0. Figures E.3 and E.4 plot 432 district by period rates each, reproduced from the deposited officer-day panel.

One caption does not describe its figure. Figures E.3 and E.4 are captioned “90 days before and after BWC deployment” and say there is no significant difference “in either the 90-day period before or after”. The published axis runs from roughly 750 days before deployment to 500 days after, and the underlying panel covers that whole span. The window in the caption is not the window in the plot.

7 Verification

Every script runs in a clean session and the whole pipeline runs end to end from `run_all.R`. Running it twice returns byte-identical CSV, LaTeX and PNG output; the four figure PDFs are the only files that change, and they change because a PDF records the time it was written. No estimator used anywhere here draws a random number, so there is no seed to set and no dependence on which sampler the running version of R provides.

`run_all.R` begins by sourcing `download_original.R`, so every run re-verifies `original/` against the manifest before any analysis reads from it, and stops if a file has changed or if a file the manifest does not list has appeared. The deposit's own scripts are never executed inside `original/`, which is what a checksum gate placed after the fact would fail to catch.

Table 9: R environment. Rendered under R version 4.6.0 (2026-04-24).

Package	Version
tidyverse	2.0.0
estimatr	1.0.6
broom	1.0.13
ggplot2	4.0.3
readr	2.2.0
dplyr	1.2.1
purrr	1.2.2
knitr	1.51
kableExtra	1.4.0
here	1.0.2
openssl	2.4.2